International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2021-0561
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First Published Online: July 20, 2022

Human Kinetics
BRIEF REPORT

A Strategy to Inform Athlete Sleep Support From Questionnaire Data and Its Application in an Elite Athlete Cohort

Haresh T. Suppiah, Paul B. Gastin, and Matthew W. Driller
Sport and Exercise Science, School of Allied Health, Human Services and Sport, La Trobe University, Melbourne, VIC, Australia

Purpose: Information from the Pittsburgh Sleep Quality Index (PSQI) and Athlete Sleep Behavior Questionnaire (ASBQ) provide the ability to identify the sleep disturbances experienced by athletes and their associated athlete-specific challenges that cause these disturbances. However, determining the appropriate support strategy to optimize the sleep habits and characteristics of large groups of athletes can be time-consuming and resource-intensive. The purpose of this study was to characterize the sleep profiles of elite athletes to optimize sleep-support strategies and present a novel R package, *AthSlpBehaviouR*, to aid practitioners with athlete sleep monitoring and support efforts. ***Methods:*** PSQI and ASBQ data were collected from a cohort of 412 elite athletes across 27 sports through an electronic survey. A *k*-means cluster analysis was employed to characterize the unique sleep-characteristic typologies based on PSQI and ASBQ component scores. ***Results:*** Three unique clusters were identified and qualitatively labeled based on the *z* scores of the PSQI components and ASBQ components: cluster 1, "high-priority; poor overall sleep characteristics + behavioral-focused support"; cluster 2, "medium-priority, sleep disturbances + routine/environment-focused support"; and cluster 3, "low-priority; acceptable sleep characteristics + general support." ***Conclusions:*** The findings of this study highlight the practical utility of an unsupervised learning approach to perform clustering on questionnaire data to inform athlete sleep-support recommendations. Practitioners can consider using the *AthSlpBehaviouR* package to adopt a similar approach in athlete sleep screening and support provision.

Keywords: PSQI, ASBQ, clustering, R package, machine learning

Sleep monitoring has a vital role in the athlete recovery process.[1] This is due to the recognized importance of sleep on basal human processes and athletic performance.[2] In addition, the need for athlete sleep monitoring is justified given the prevalence of poor or inadequate sleep in developmental and elite athletes alike.[3,4] While several tools of varying validity and reliability are available for athlete sleep assessment, sleep questionnaires are often used at the onset due to their expediency in administration and low cost.[1] Furthermore, sleep questionnaires are a practical alternative for use in large-scale sleep observations in athlete populations.[3–6]

Questionnaire data on the sleep characteristics and associated presleep hygiene practices of athletes can provide practitioners with a better understanding of modifiable behaviors that require remediation or to prioritize support for athletes with less than satisfactory sleep habits. This approach has been employed elsewhere[6] to provide a more time and resource-efficient solution to guide decision making and recommendations for sleep support interventions in elite youth athletes. In our previous research, we performed an unsupervised machine learning approach to cluster athletes by sleep characteristics using the component scores of the Pittsburgh Sleep Quality Index (PSQI).[6] The use of PSQI component scores to guide decision making and recommendations for sleep interventions has been acknowledged.[7] This approach has also been recognized to benefit from additional athlete-specific information through questionnaires such as the Athlete Sleep Behavior Questionnaire (ASBQ),[7] for targeted behavioral strategies to improve sleep.[5] Thus, an approach that combines information from the PSQI and ASBQ may provide practitioners with a more nuanced appreciation of the potential sleep disturbances faced by athletes, as well as the athlete-specific challenges that may cause these disturbances.

While studies have highlighted the prevalence of poor sleep duration and quality, and suboptimal sleep hygiene behaviors in athletes, there is a lack of implementation of sleep monitoring processes among practitioners.[8] Despite them having an appreciation of the importance of sleep, a lack of resources has been attributed to nonimplementation.[8] Translating sleep monitoring data to implementable decisions on how to address these sleep issues is further hampered by the abundance of data, potentially unoptimized data workflow, and time limitations faced by sport science support teams.[9]

Herein, we present a case study on the use of PSQI and ASBQ data to characterize the sleep profiles of elite athletes to optimize sleep support strategies. To allow practitioners to implement a similar sleep support strategy for their athletes, and, based on recent recommendations to bridge the gap from research to practice,[10] we also make available a novel R package, *AthSlpBehaviouR*, which aids with the retrieving, scoring, modeling, visualization, and deployment of PSQI and ASBQ data.

Methods

Data Collection

In a cross-sectional design, a convenience sample of 412 elite athletes across 27 sports (Table 1) completed the PSQI and ASBQ via an electronic survey (Survey Monkey, Palo Alto Inc). Participants were recruited via a combination of social media channels, word of mouth, and targeting specific sporting organizations and national bodies. The criteria for inclusion in the current study was representation at either national or international level for their chosen sport.

Suppiah (h.suppiah@latrobe.edu.au) is corresponding author.

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Table 1 Participant Demographics

Variable	Male, n = 263[a]	Female, n = 149[a]
Age, y	23.2 (4.2)	23.0 (4.3)
Sport		
Rugby union	149 (57%)	61 (41%)
Hockey	12 (4.6%)	16 (11%)
Rowing	10 (3.8%)	17 (11%)
Rugby league	21 (8.0%)	0 (0%)
Basketball	8 (3.0%)	7 (4.7%)
Cricket	15 (5.7%)	0 (0%)
Athletics	7 (2.7%)	7 (4.7%)
Netball	0 (0%)	12 (8.1%)
Cycling	8 (3.0%)	4 (2.7%)
Football	9 (3.4%)	1 (0.7%)
Swimming	3 (1.1%)	6 (4.0%)
Tennis	7 (2.7%)	2 (1.3%)
Triathlon	3 (1.1%)	4 (2.7%)
Golf	3 (1.1%)	4 (2.7%)
Volleyball	1 (0.4%)	2 (1.3%)
Waterskiing	0 (0%)	2 (1.3%)
Boxing	1 (0.4%)	0 (0%)
Touch rugby	1 (0.4%)	0 (0%)
Futsal/football	1 (0.4%)	0 (0%)
Ice hockey	0 (0%)	1 (0.7%)
Kayaking	1 (0.4%)	0 (0%)
Badminton	0 (0%)	1 (0.7%)
BMX racing	1 (0.4%)	0 (0%)
Canoe sprint	0 (0%)	1 (0.7%)
Show jumping	0 (0%)	1 (0.7%)
Clay target shooting	1 (0.4%)	0 (0%)
Track cycling–sprint	1 (0.4%)	0 (0%)
PSQI global score	5.16 (2.74)	6.12 (2.68)
ASBQ global score	44 (6)	43 (6)

Abbreviations: ASBQ, Athlete Sleep Behavior Questionnaire; PSQI, Pittsburgh Sleep Quality Index.
[a]Mean (SD) or mean n (percentage).

Statistical Analysis

All statistical analyses were done using R (version 4.1.1, R Core Development Team). A *k*-means cluster analysis was conducted to identify unique subgroups of sleep characteristics elicited from the PSQI and ASBQ components. *K* means is an unsupervised machine clustering algorithm to partitioning data that makes no a priori assumptions about the data. The algorithm is an iterative technique that attempts to partition an unlabeled data set into *K* unique nonoverlapping subgroups (ie, *K* clusters), with each data point assigned to a cluster. In principle, the algorithm classifies objects based on information from the data that explains the relationship between objects and does so by assigning data points to a cluster such that the sum of their squared distances from a cluster's centroid is minimized.[11] The elbow method, which uses the sum of squared distances to determine the optimal number of *K* clusters based on the distance between the data points and their assigned clusters, was used to determine the ideal value of *K*.[12] Variables were transformed into standardized *z* scores to standardize the scaling across the variables prior to clustering.

R Package Development

The *AthSlpBehaviouR* R package was developed using R (version 4.1.1, R Core Development Team) to retrieve, score, model, and visualize ASBQ and PSQI data collected on a Google Form. At the time of writing, the package includes 2 major functions to automatically (1) clean and score data and (2) create a dashboard to aid in the categorization of sleep characteristics and behaviors using a *k*-means cluster analysis. Source code, documentation, and pseudo data pertaining to the package are available in Supplementary Material S1 (available online). The supplementary material also includes a detailed guide to implementation, and a downloadable Google Form template of the PSQI and ASBQ instruments.

Results

A 3-cluster solution was determined to be the optimal number of clusters based on the elbow method (Figure 1).[11] The sleep characteristics of each cluster were qualitatively labeled according to the most prominent behaviors in each cluster (Figure 2). Based on the *z* scores of the PSQI components and ASBQ components, the clusters were qualitatively labeled, cluster 1: "high-priority; poor overall sleep characteristics + behavioral-focused support," cluster 2: "medium-priority; sleep disturbances + routine/environment-focused support," and cluster 3: "low-priority; acceptable sleep characteristics + general support." A detailed breakdown of the cluster characteristics by sport is provided in Supplementary Material S2 (available online).

Discussion

The aim of this report was to demonstrate the use of PSQI and ASBQ data to characterize athlete sleep profiles to optimize sleep support strategies. The application of the *k*-means unsupervised machine learning algorithm on these data can help with the identification of latent characteristics. The application of this approach in the present study identified 3 possible clusters, all of which display unique characteristics and may inform individualized approaches to sleep recommendations. Sport scientists and practitioners can consider the information derived from this approach for more targeted support in improving athlete sleep.

With questionnaires allowing for the collection of large datasets on athlete sleep characteristics, there is value in using an unsupervised clustering algorithm to assist practitioners with mining the data and differentiate sleep typologies based on the grouping patterns of the data. The 3 identified clusters based on the PSQI and ASBQ data facilitate the prioritization and planning of sleep strategies optimal for each subgroup. For instance, athletes within cluster 1 presented with higher scores in nearly all PSQI components and generally higher scores on ASBQ behavioral factors and therefore qualitatively labeled, "high-priority; poor overall sleep characteristics + behavioral-focused support." Based on the characteristics of this group, a support strategy might be to prioritize objective sleep monitoring and the use of cognitive behavior therapy to alleviate the presleep anxiety concerns of athletes in this cluster.[1] Although only a relatively small proportion (14.8%) of athletes were categorized in this cluster, the sharp contrast of their PSQI-obtained *sleep duration* and *sleep efficiency*

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Figure 1 — UMAP (uniform manifold approximation and projection) 2-dimensional projection with *k*-means cluster assignment. Cluster 1—high priority, poor overall sleep characteristics + behavioral-focused support; cluster 2—medium priority, sleep disturbances + routine/environment-focused support; and cluster 3—low priority, acceptable sleep characteristics + general support.

component scores to those of the other 2 groups are noteworthy. Previous research[13] has highlighted that athlete sleep durations and sleep efficiency worsen during periods of heavy training, which may warrant further investigation and intervention within the athletes' training schedules. Within a multidisciplinary sport science team, education sessions by a sports scientist on optimal rehydration strategies during the day and strategies to reduce delayed onset muscle soreness symptoms might also be useful preliminary strategies to assist with facilitating improved sleep in this group of athletes.[14] Athletes within cluster 2 (medium-priority; sleep disturbances + routine/environment-focused support) might be monitored through daily sleep diaries, as well as interviewed, to understand reasons behind bedtime and waketime inconsistency to determine if these were due to controllable factors by the athlete (eg, device usage) or externally influenced (eg, training/competition scheduling). Depending on the reasons behind sleep variability, sleep education would have to be appropriately provided to athletes or coaches/administrators for best effect. Finally,

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Variable	Cluster 1, N = 61	Cluster 2, N = 163	Cluster 3, N = 188
ASBQ			
Routine/ environmental factors[a]			
Afternoon nap	0.09	0.10	−0.12
Bedtime variation	0.28	0.39	−0.43
Wake-time variation	0.07	0.34	−0.32
Poor sleep environment	0.39	0.27	−0.36
Foreign sleep environment	−0.12	0.21	−0.15
Constant travel	0.43	0.14	−0.26
Behavioral factors[a]			
Stimulant use	0.39	0.06	−0.18
Alcohol consumption	0.26	0.21	−0.27
Light exposure	0.13	0.14	−0.16
Non-sport-related concerns	0.83	0.24	−0.48
Use of sleeping pills	0.26	0.21	−0.27
Bathroom use	0.10	0.24	−0.24
Snoring	0.28	0.19	−0.26
Sport-related factors[a]			
Night training/competition	0.24	0.31	−0.35
Thirst	0.48	0.28	−0.40
Muscle soreness	0.42	0.38	−0.47
Concerns (sport related)	0.80	0.26	−0.48
Muscle twitching	0.13	0.42	−0.41
PSQI components			
Sleep quality	0.99	0.10	−0.4
Sleep latency	0.85	0.10	−0.36
Sleep duration	1.51	−0.26	−0.26
Sleep efficiency	1.52	−0.27	−0.26
Sleep disturbances	0.77	0.23	−0.45
Sleep medication	0.24	0.13	−0.20
Daytime dysfunction	0.59	0.17	−0.34

Figure 2 — Mean final cluster centers (standardized *Z* values) of ASBQ and PSQI variables. ASBQ indicates Athlete Sleep Behavior Questionnaire; PSQI, Pittsburgh Sleep Quality Index. A red-to-green color scale is implemented on *Z*-score values to assist with the visualization of the data distribution. (Refer to the online color version of the article.) Red-highlighted variables indicate higher *Z*-score values and ones of concern in relation to sleep behavioral disturbances, while green-highlighted variables are indicative of lower *Z*-score values and of lower concern in relation to sleep behavioral disturbances. [a]Factors based on previously identified underlying dimensions.[7]

athletes belonging to cluster 3 (low-priority; acceptable sleep characteristics + general support) may benefit from advice to maintain their current sleep habits with periodic questionnaire-based reassessments of their sleep characteristics for better prioritization of practitioner time and sleep monitoring resources.

To provide practitioners with the ability to obtain similar actionable insights, we also present the *AthSlpBehaviouR* R package to streamline the retrieval, scoring, modeling, and visualization of ASBQ and PSQI data. This resource aims to address and provide practitioners with a coherent and coordinated data workflow from the point of data acquisition to decisional guidance on sleep support strategies. The embedded *k*-means clustering function assists in grouping athletes with similar attributes based on the ASBQ and PSQI, which may be challenging to interpret manually due to the number of variables and potential size of the athlete cohort. In this regard, the usefulness of the package would likely be in scenarios in which data from large numbers of athletes across multiple sports might be collected (eg, youth academies, sport schools, and college squads). The development of this package was aimed to provide practitioners with a research-informed tool to optimize athlete sleep habits.[10] Furthermore, we suggest that adopting a group-specific approach to sleep recommendations and support could be a more practically feasible and efficient approach for practitioners working with large athletes.

Practical Applications

Clustering techniques can be considered by applied sport practitioners to better identify the necessary sleep support strategies for large athlete squads for optimal recovery outcomes. Automating the workflow of data acquisition to insight generation of athlete sleep behaviors and characteristics through the *AthSlpBehaviouR* package could facilitate the adoption of sleep monitoring among sleep practitioners. This approach allows practitioners to identify and focus on athletes that are at greater risk of poor sleep quality and behaviors.

Conclusions

The strategies and applications of a clustering technique outlined in this study provide practitioners with an approach to prioritize and provide targeted information and advice to athletes based on similar (ie, grouped) behaviors, routines, and environments. This may be particularly useful in the applied setting when dealing with large groups/squads of athletes where it may not be possible to provide individual sleep advice based on the results from sleep questionnaires.

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